



Microprocessor and Interfacing Tut-5

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Q1. You have to write a macro named **NINE** which performs the following operation **OutPar=9*InPar** where **InPar** and **OutPar** are the parameters to the macro **NINE**. You are not allowed to use MUL/IMUL in this macro. You are allowed to use **ONLY CX** inside this macro, and are allowed to use only one ADD instruction. Assume that the range of InPar is from 50H – 200H. ***The macro should not have more than 6 lines of code (including the macro header and footer).***

Furthermore, write a complete 8086 ALP which will change the contents of **DX** register with **27** times the content of **BX** (i.e., **DX=27*BX**). This ALP is allowed to call the macro **NINE** one time. **Only BX, DX and CX** can be used in this program. ***The code between. startup and. exit should not exceed 4 lines.*** You can use the ADD instruction only once in the ALP. No other arithmetic instruction can be used.

Write the same code using procedures parameters as labels



Q2. You have to write a procedure named **NINE** which performs the following operation **OutPar=9*InPar** where **InPar** and **OutPar** are the **labels** to **word-sized** memory locations. You are not allowed to use MUL/IMUL inside the procedure. You are allowed to use **ONLY AX, CX** inside this procedure, and are allowed to use only one ADD instruction. However contents of AX,CX should be unaltered after the procedure. Assume that the range of InPar is from 50H – 200H. ***The procedure should not have more than 8 lines of code (including the procedure header and footer).***

Write the same code using procedures parameters as pointers



Q3. You have to write a procedure named **NINE** which performs the following operation **OutPar=9*InPar** where **InPar** and **OutPar** are the **labels** to **word-sized** memory locations. The offsets of these locations are loaded in SI (for InPar) and DI (for OutPar) You are not allowed to use MUL/IMUL inside the procedure. You are allowed to use **ONLY AX, CX** inside this procedure, and are allowed to use only one ADD instruction. However contents of AX,CX should be unaltered after the procedure. Assume that the range of InPar is from 50H – 200H. ***The procedure should not have more than 9 lines of code (including the procedure header and footer).***

Q4. Two word-sized arrays Dat1 & Dat2 have 50 elements each. 49 elements of the arrays are same and only 1 element is different. Write an 8086 ALP to find the index of the(differing) element and store the index in the location labelled '**Index**' after the code is executed.

For this example, you can assume that the index starts from 1. For example, the index of 67h in Dat1 is 1, the index of 66h is 2 and so on.

Dat1 dw 67h, 66h, 70h, 40h.....